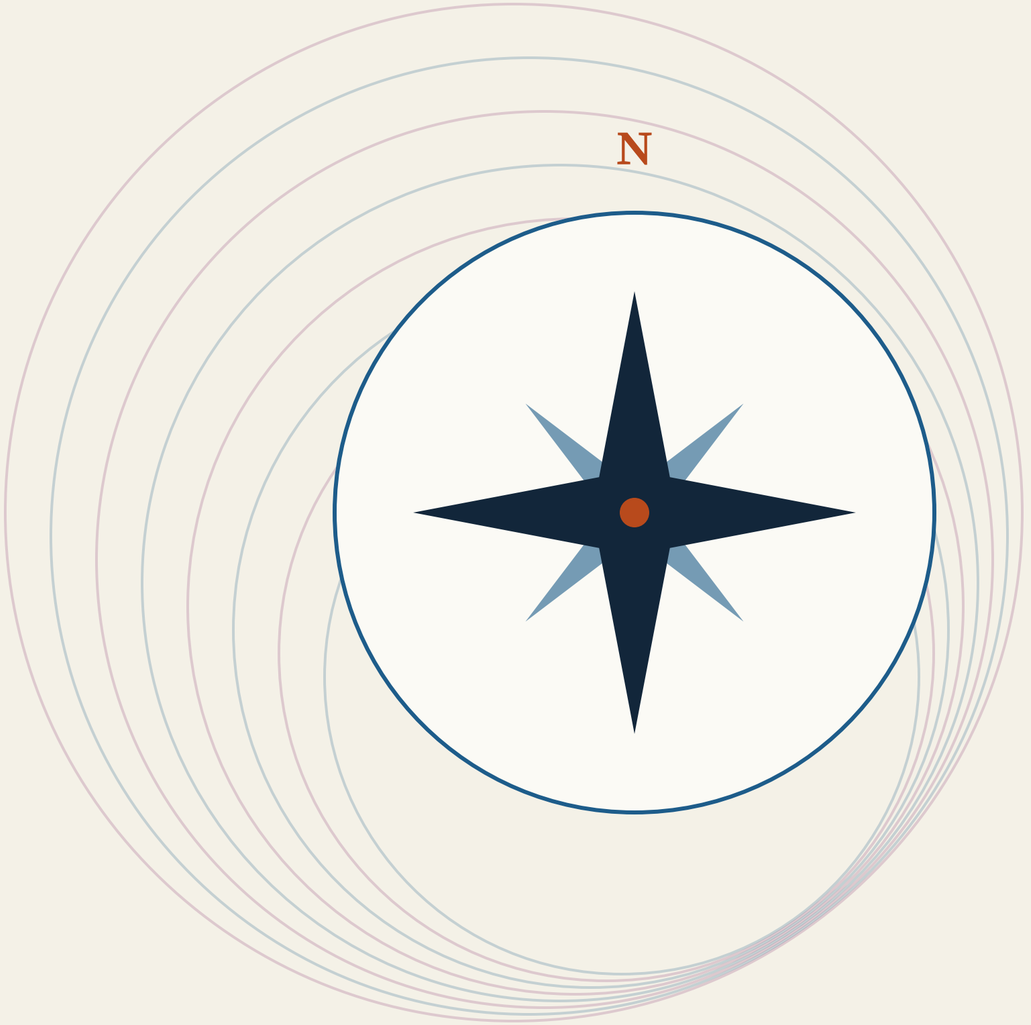
SkillTide

See where AI is taking your career before you graduate.



A career navigator for the AI era
Team Traiblazers · Xiamen University Malaysia
SDC Hackathon 2026 · Domain 1: The Aiversecity Future-Ready Workforce
SDG 4: Quality Education · Live at skilltide.replit.app

AI is changing entry-level tech jobs faster than degrees and career advice can follow

AI use at work has outpaced training, and E&E and ICT, where our students are headed, are directly affected.

56%

Use AI tools at work

of Malaysian professionals already do, but only 26% have been trained to use them.

Hays Asia Salary Guide, Mar 2025

697,000

Workers affected

in 22 sectors will be significantly affected by AI, digitalisation and the green economy within 3 to 5 years.

TalentCorp, via Bernama, 1 May 2026

37,000

E&E jobs affected

and 30,000 ICT jobs, in TalentCorp's first impact study: the sectors our students are about to enter.

The Star, 19 Aug 2025

−11.4%

Online job offerings

in a single quarter, down to 555,509 in Q2 2026.

DOSM, Job Market Insights Q2 2026

Final-year students face three gaps

For final-year students and fresh graduates in computer science, electrical & electronic engineering and mechatronics in Malaysia — 6 to 12 months before the job hunt.

Gap 1 · No live view of change

Skill and job lists are out of date by the time they are published.

SkillTide: dated public signals, refreshed by a tested data pipeline.

Gap 2 · No evidence behind advice

A general chatbot cites nothing and can invent jobs, courses or numbers.

SkillTide: every role, course and number links to its source.

Gap 3 · No plan that fits final year

A few hours a week, little money and a fixed deadline.

SkillTide: a week-by-week plan of verified free or low-cost courses.

SkillTide in four steps, with a Guide beside every answer

Live at skilltide.replit.app, in English, Bahasa Melayu and Simplified Chinese.

1 · Profile

Paste or upload a CV, tick skills or answer quick questions. The CV is read in the browser and never uploaded.

2 · Skills Radar

57 skills and 30 roles marked rising, steady, transforming or under pressure, from dated data.

3 · Matches

Roles ranked by a visible score: skills fit, AI outlook and SDG fit, with weights the student sets.

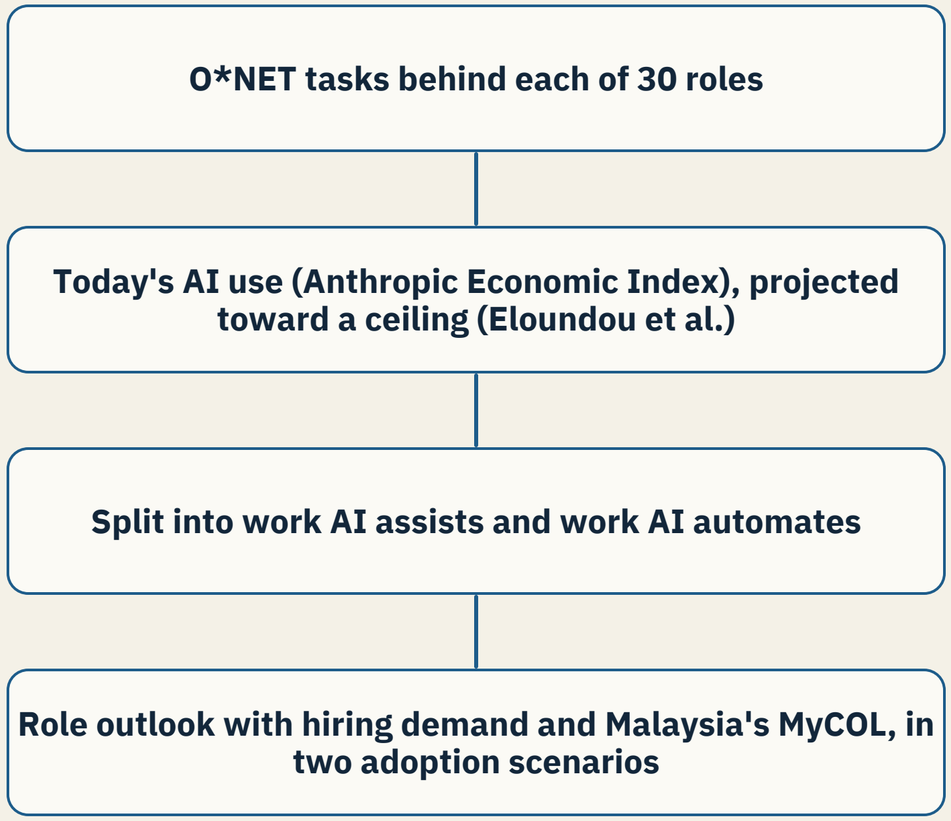
4 · Pathway

Verified free or low-cost courses, week by week, ending in a portfolio project built on a real task of the role.

Guide · answers “why?”, “what if?” and “where from?” with sources, and re-plans when the data or scenario changes

Innovation: forecasting AI task by task, not by job title

657 of the 658 O*NET tasks behind 30 roles get their own AI-exposure estimate, which then rolls up into roles and skills.



TASKS BEFORE TITLES

Exposure after h = 3 years:
$e(h) = c - (c - e_0)(1 - r)^h$
e_0 today's use · c ceiling · r adoption rate (15% Moderate, 35% Fast)

Outlook = 50 + 20·demand + 40·assisted – 55·automated, +5 if on MyCOL

Six dated snapshots can be replayed to see how the picture changed.

Demo: Kavin, a synthetic mechatronics student in Johor

One student's decision, step by step, with real outputs from the live app. Kavin is synthetic: no real CVs are used.

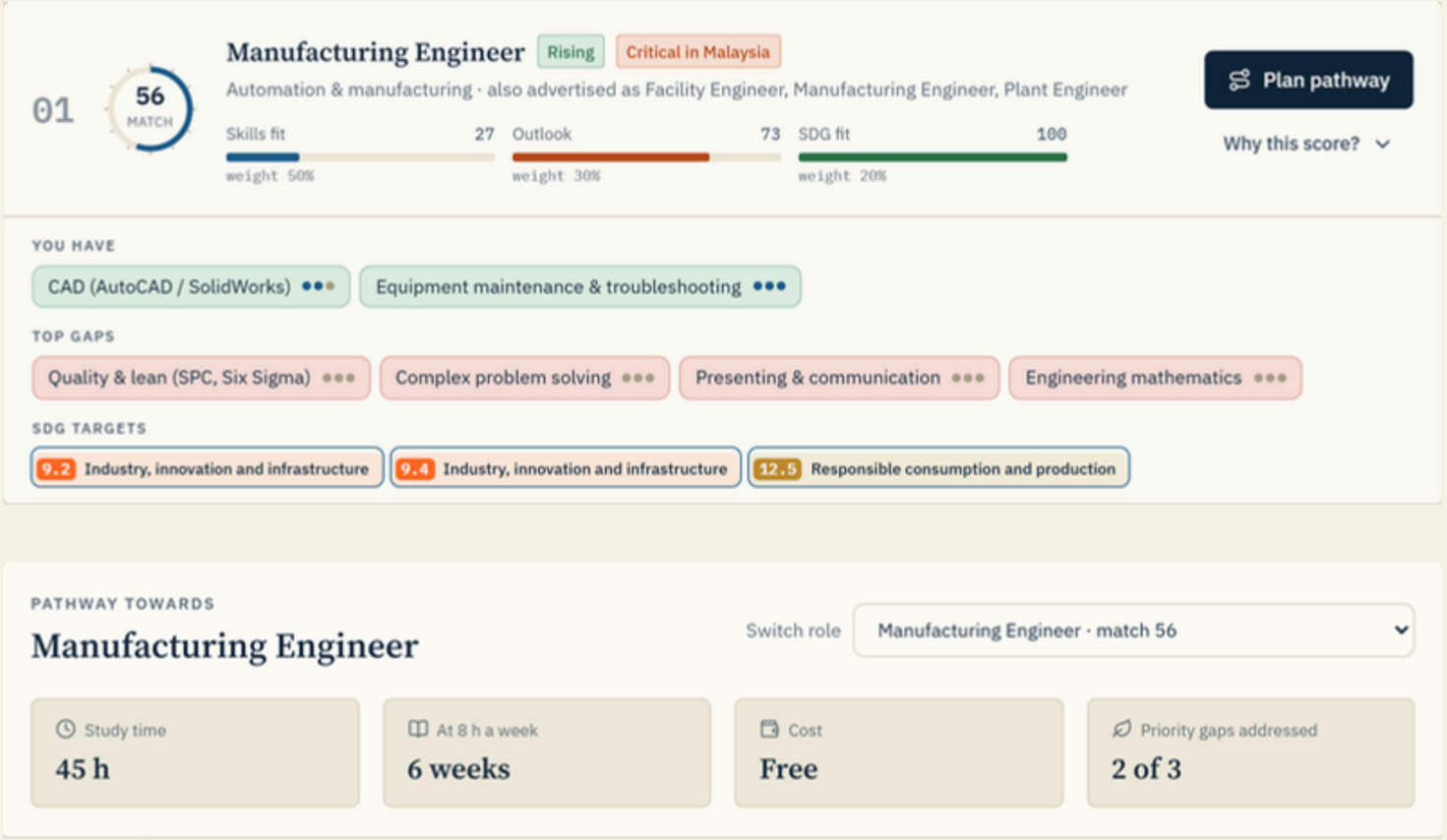
1 His CV yields 12 skills, each graded by where it appears: internship, projects or coursework.

2 He picks SDG 9 and SDG 12, and Manufacturing Engineer moves to first place with a match of 56.

3 With March 2025 data, the same profile pointed to Electro-Mechanical & Mechatronics Technician.

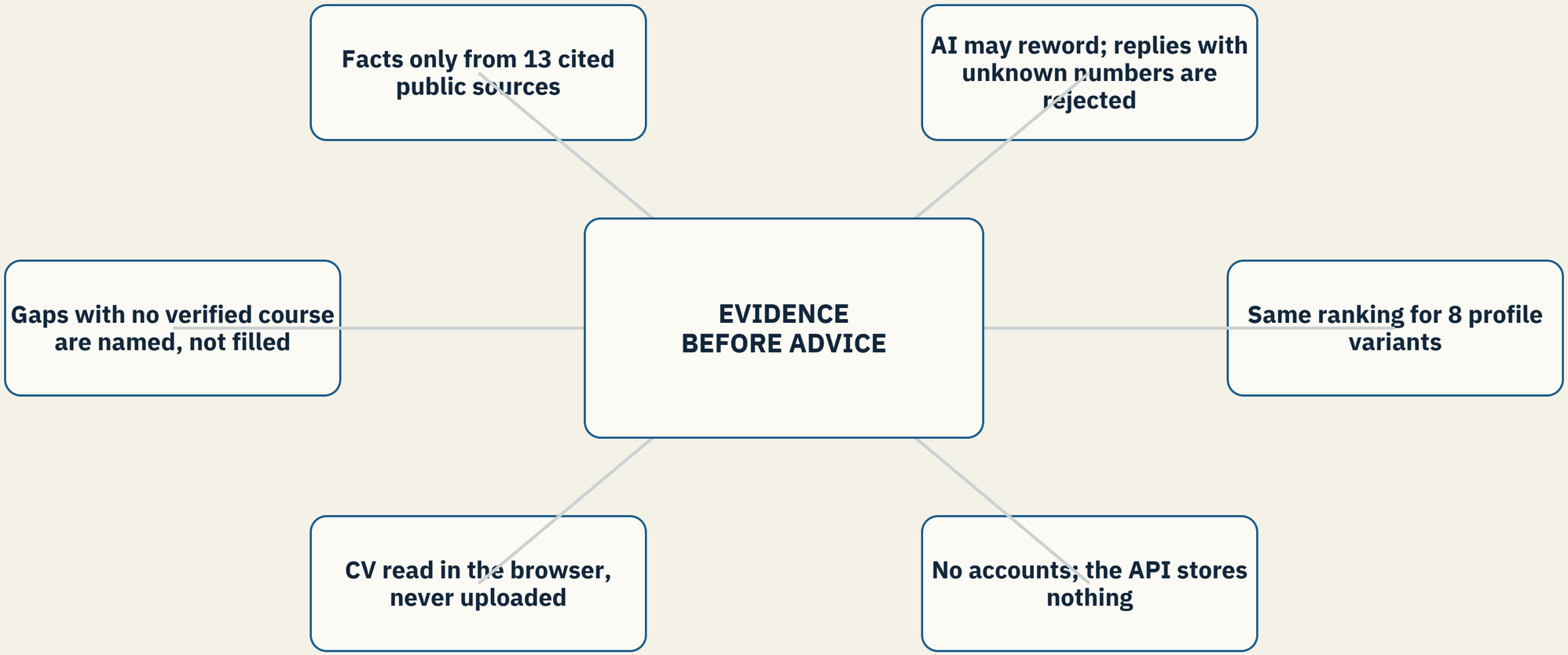
4 Under fast AI adoption it still leads, but its outlook drops from 73 to 67.

5 His plan: 45 hours over 6 weeks at 8 hours a week, two free courses and a 20-hour portfolio project.



Trust by design

Students and career centres can inspect how an answer was produced — not merely accept a score.



Scoring never uses names or backgrounds. Replays use today's catalogue, so they are not a backtest. Twelve synthetic test cases check coverage, budgets and fairness: design checks, not proof of job outcomes.

For career centres: the same engine across a whole cohort

The same engine, run across a year group: where rising skills are thin, where to run workshops, and which SDG targets the cohort is heading towards.

180

Synthetic final-year students

60 each in computer science, electrical & electronic and mechatronics, drawn from typical skill mixes with a fixed seed.

A real career centre would load its own anonymised, aggregated data instead.

Demo cohort, not real students. 180 synthetic final-year students (60 per programme) drawn from typical skill mixes with a fixed seed. A real career centre would load its own anonymised, aggregated survey data instead.

Moderate AI · data Latest (16 Oct 2026)

Skills heatmap

Share of each programme with the skill at project level or above, for the 16 technical skills with the strongest outlook

Skill	Outlook	Computing	Electrical & electronic	Mechatronics
CAD (AutoCAD / SolidWorks)	Rising	0%	0%	53%
Digital design (Verilog / VHDL / FPGA)	Rising	0%	22%	0%
Energy efficiency & audits	Rising	0%	12%	0%
C / C++	Rising	30%	40%	38%
Solar PV design	Rising	0%	5%	0%
Signal processing	Rising	0%	40%	0%
Mobile apps (Android / iOS)	Rising	12%	0%	0%
Embedded systems & microcontrollers	Rising	0%	28%	32%
MATLAB & Simulink	Rising	0%	47%	32%

Where to run workshops

Rising skills the fewest students have at project level.

C# / .NET
Outlook score 74, but only 0% of the cohort can use it in a project.

Solar PV design
Outlook score 78, but only 2% of the cohort can use it in a project.

Semiconductor & electronics testing
Outlook score 62, but only 2% of the cohort can use it in a project.

Energy efficiency & audits
Outlook score 80, but only 4% of the cohort can use it in a project.

Impact: measured today, with targets for Phase 2

Linked to SDG target 4.4: more young people with relevant skills for decent jobs.

How we measure

180

synthetic final-year students
Data of 16 Oct 2026
Moderate AI scenario

Anyone can reproduce it with npm run impact

78% → 90%

Gap coverage: priority skill gaps closed by a verified free or low-cost course. Today 696 of 889; Phase 2 target at least 90%.

28% → 50%

Fully covered students: can close every priority gap. Today 51 of 180; Phase 2 target at least 50%.

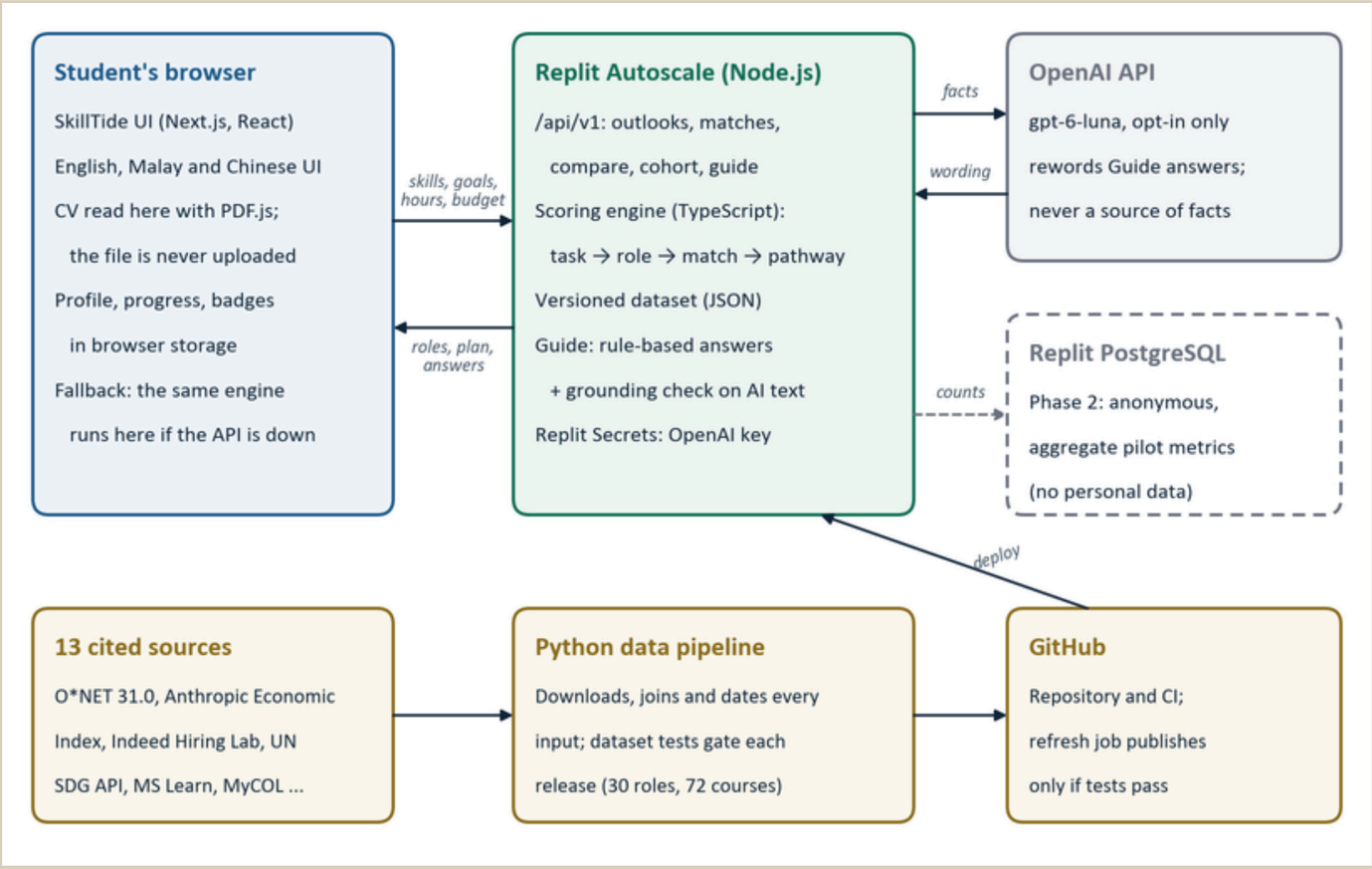
≥ 80%

Pilot usability: share of 10 or more final-year volunteers who reach a first pathway in under 5 minutes and can explain their top role's score. Measured in Phase 2.

Most open gaps are in equipment maintenance, lean and quality, and MATLAB, which have no verified free course yet. Phase 2 verifies and adds free courses for them.

Feasibility: live, tested and cheap to run

One TypeScript engine on the server, or in the browser as a fallback; data rebuilt from 13 public sources.



LIVE

Replit Autoscale (Asia) at skilltide.replit.app, plus a Cloudflare mirror. If the API is down, the browser runs the same engine.

TESTED

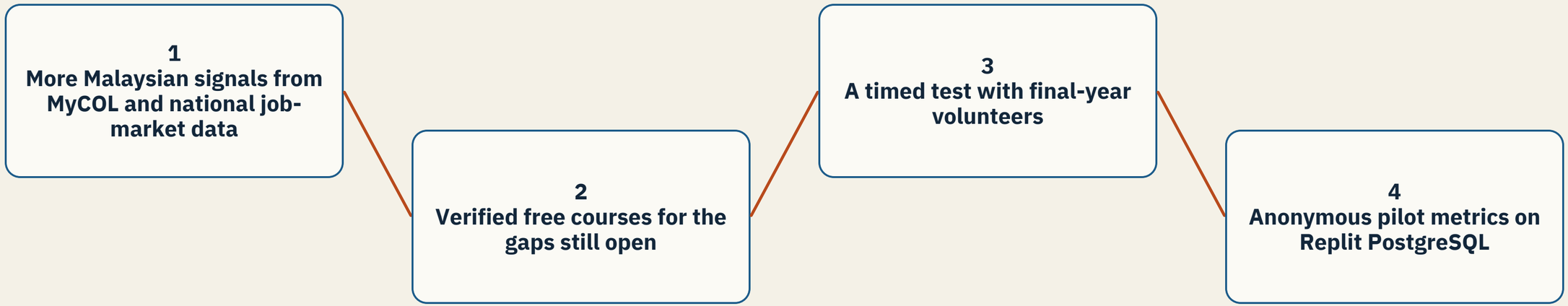
106 automated tests run in CI, plus a 12-case synthetic evaluation. A data refresh publishes only if the tests pass.

CHEAP TO RUN

Public data and free courses. Works fully without an AI key; the optional AI wording uses our own OpenAI key.

Next: Phase 2 and the final

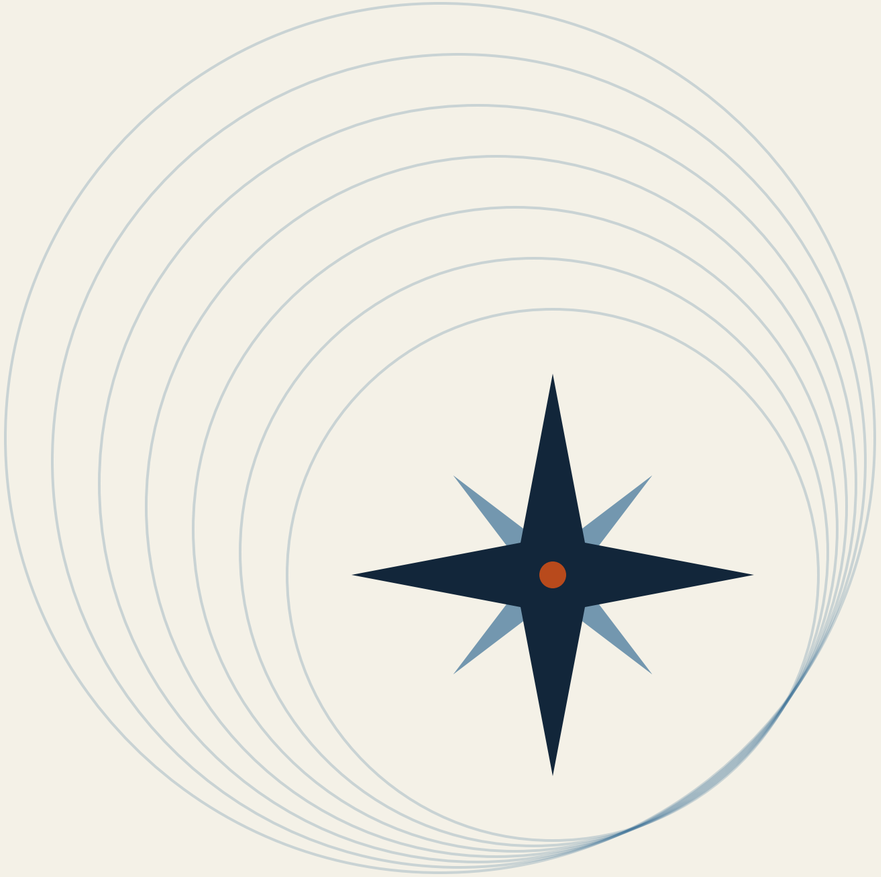
If we are shortlisted: Phase 2 online (28 to 30 Oct), then the 24-hour final at XMUM (31 Oct to 1 Nov).



Also built: a Bahasa Melayu and Chinese interface, progress badges and study circles, and opt-in AI wording checked against the data.

Chart your course before you graduate

Team Traiblazers · Xiamen University Malaysia
HUANG BAONING (team leader) · WANG YUZHE



Live app: skilltide.replit.app

Code: github.com/ArcueidMP/skilltide

5-minute demo film: linked in the repository README

Declared AI use: the team set the users, scope, data rules and design; Claude Code and Codex wrote code under our direction and review.